

CEL: controlled comparisons, visible trade-offs

Hold the protocol constant. See what each method actually trades.

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Code & project

BENCHMARK SCOPE

One broad benchmark

18
datasets

14
methods

3
paradigms

2
backbones / task

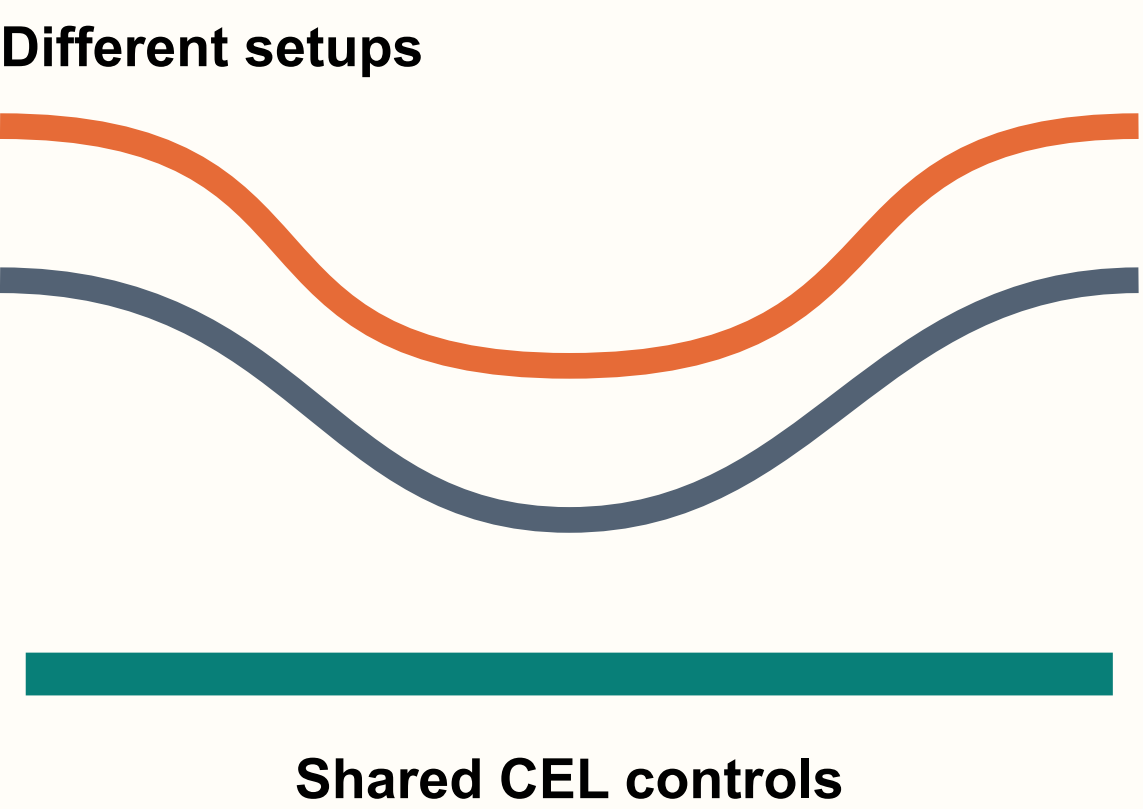
5
folds

Source · Datasets · Methods · Full-Results

LEFT

Protocol differences can confound comparisons

Splits, preprocessing, predictors, constraints, and metrics can obscure method behavior.



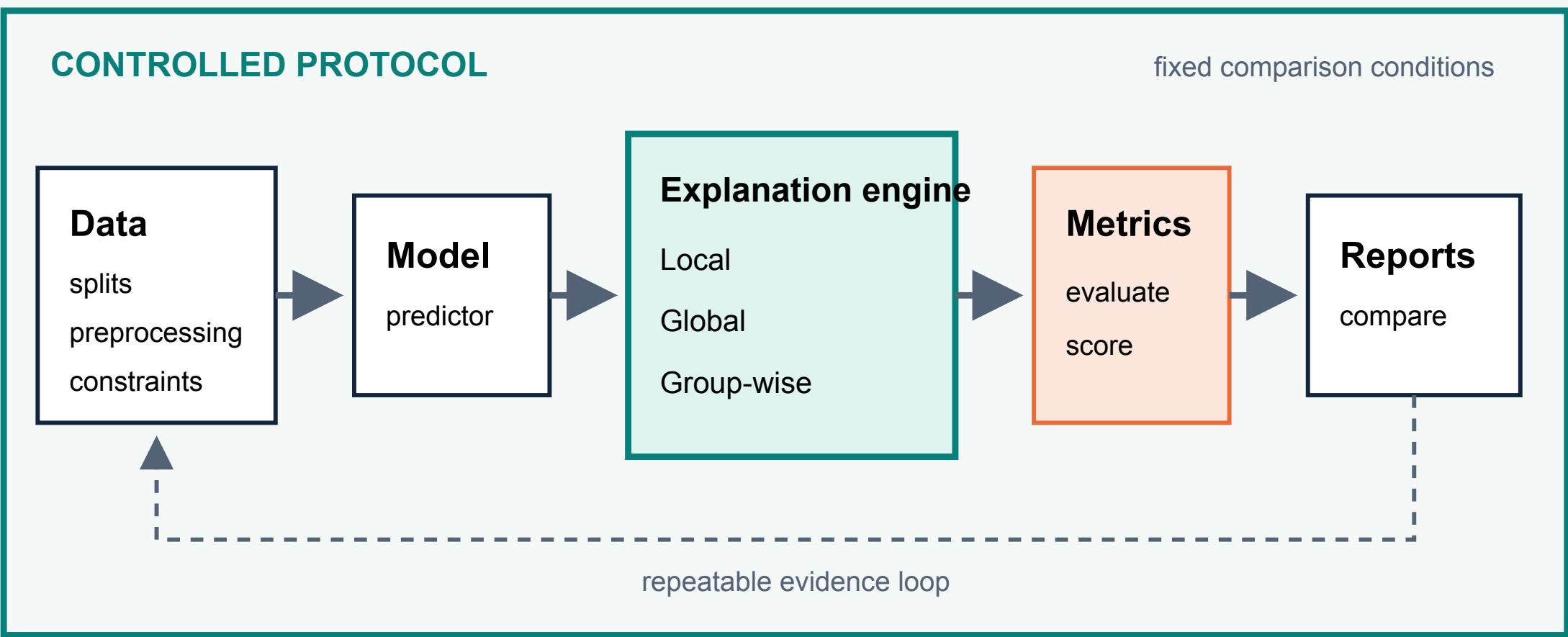
CEL reduces experimental setup as a confounder in method comparisons.

Source · Benchmark

CENTER

CEL standardizes a shared protocol

Splits, preprocessing, predictors, constraints, and metrics follow shared benchmark controls.



ONE CONTROLLED PROTOCOL: FIXED SPLITS, PREPROCESSING, PREDICTORS, CONSTRAINTS, AND METRICS

CEL reduces experimental setup as a confounder in method comparisons. Counterfactual quality is a trade-off among success, change size, plausibility, applicability, and runtime.

Source · Benchmark

RIGHT

Plausibility can cost more change

Read plausibility together with perturbation size inside one named configuration.

BLOBS · LOCAL · MLP

PPCEF plausibility higher plausibility ↑ 1.00 ± 0.00

PPCEF distance lower change ↓ 0.48 ± 0.03

PPCEF's displayed distance (0.48 ± 0.03) is larger than the row-group minimum, 0.34 ± 0.03 for CADEX.

Source · tab:num_metrics_mlp

RIGHT

Shared shifts trade reach for size

Higher group-wise effectiveness accompanies a larger shared change in the selected Adult comparison.

ADULT CENSUS · GROUP-WISE · MLP

method	validity ↑	change ↓
T-CREx	0.23 ± 0.02	0.00 ± 0.00
GLANCE	0.99 ± 0.01	0.41 ± 0.02

Adult Census, MLP. Reported as 0.00 ± 0.00 after rounding; not asserted to be an exact zero.

Source · tab:group_metrics_mlp

BOTTOM

Similar targets, different perturbations

On Concrete, target error, change size, and plausibility tell different parts of the comparison.

CONCRETE · REGRESSION · MLP

metric	CEARM	Wachter
Target MAE ↓	0.04 ± 0.01	0.03 ± 0.00
L2 change ↓	1.16 ± 0.02	0.09 ± 0.00

Plausibility: 0.24 ± 0.02 Wachter's displayed plausibility mean (0.24 ± 0.02) is higher than CEARM's (0.00 ± 0.00).

Source · tab:regression_all_dnn

BOTTOM

Missing output is part of the evidence

An unreported result is neither zero nor proof of method inapplicability.

BLOBS · GLOBAL · MLP



Blobs, MLP. The source reports a missing-value marker; this is missing output, not proof of method inapplicability.

Source · tab:global_metrics_mlp

BOTTOM

Choose by task, metric, and applicability

Inspect coverage; validity can depend on result semantics.

Compare plausibility within one dataset and model. Rounded zeroes are not exact.

Sparsity comparison is omitted until its direction is reconciled.

Need reliable success? Inspect validity after coverage.	Need small change? Read distance with rounded-zero semantics.
Need plausibility? Compare inside one dataset and model.	Missing output? Keep missing distinct from zero or inapplicable.

Counterfactual quality is a trade-off among success, change size, plausibility, applicability, and runtime.

Resolution: omit sparsity comparisons from the poster; retain only the caveat in the research ledger.

Source · Conclusions · tab:cat_metrics_mlp · tab:num_metrics_mlp · tab:group_metrics_mlp · tab:global_metrics_mlp · Metrics